Align Filter and Non-Holonomic Mount Observations

sensor_fusion documentation

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Contents

1 Scope

This document records the mathematical model implemented by `sensor_fusion/src/align/mod.rs`. The align filter is a standalone IMU-to-vehicle mount estimator. It estimates the mount quaternion and its three-dimensional small-angle covariance. The runtime fusion pipeline uses this mount as the coarse seed for EKF initialization.

2 Frames and Quaternion Convention

Symbol	Meaning
n	Local navigation frame, NED: north, east, down. GNSS velocities supplied to align are in this frame.
b	Raw IMU body frame. Mean gyroscope and accelerometer samples are expressed in this frame.
v	Vehicle frame, FRD: forward, right, down. The non-holonomic assumption constrains lateral and vertical vehicle velocity.

The project math notes use active rotation matrices

$$x_a = C_{ab} x_b, \tag{1}$$

where C_{ab} maps a vector from frame b into frame a. Quaternions use the same direction:

$$R(q_{ab}) = C_{ab}, \qquad R(q_1 \otimes q_2) = R(q_1)R(q_2). \tag{2}$$

The align state stores the public physical vehicle-to-body mount q_{bv}. Its rotation matrix is $C_{bv} = R(q_{bv})$:

$$x_b = C_{bv} x_v. \tag{3}$$

The body/sensor-to-vehicle rotation is the transpose:

$$x_v = C_{vb} x_b = C_{bv}^T x_b. \tag{4}$$

For small-angle corrections, the implementation uses

$$\delta q(\delta\theta) \simeq \begin{bmatrix} 1 & \frac{1}{2}\delta\theta^T \end{bmatrix}^T. \tag{5}$$

General roll/pitch/yaw mount corrections are left-injected,

$$q_{bv}^{+} = \delta q(\delta\theta) \otimes q_{bv}^{-}, \tag{6}$$

while the horizontal heading update applies a vehicle-frame yaw correction on the right,

$$q_{bv}^{+} = q_{bv}^{-} \otimes q_z(\delta\psi_v). \tag{7}$$

3 State and Covariance

The nominal state is only the mount quaternion:

$$x = q_{bv} \in \mathbb{S}^3. \tag{8}$$

The covariance is defined on a three-state small-angle error:

$$\delta x = \begin{bmatrix} \delta\phi & \delta\theta & \delta\psi \end{bmatrix}^T \in \mathbb{R}^3, \qquad P \in \mathbb{R}^{3\times 3}. \tag{9}$$

The state order is roll, pitch, yaw of the mount error. Standard-deviation configuration fields are one-sigma values; the filter stores variances in P.

4 Stationary Initialization

Given stationary accelerometer samples $f_{b,k}$, the bootstrap mean is

$$\bar{f}_b = \frac{1}{N} \sum_{k=1}^{N} f_{b,k}. \tag{10}$$

At rest, the measured specific force is approximately opposite vehicle down, so the vehicle down axis expressed in body coordinates is initialized as

$$\hat{z}_v^b = -\frac{\bar{f}_b}{\|\bar{f}_b\|}. \tag{11}$$

The forward axis is chosen by projecting a reference body x-axis into the plane orthogonal to $\hat{z}_v^b$:

$$\tilde{x}_v^b = x_{\mathrm{ref}} - \hat{z}_v^b (\hat{z}_v^b)^T x_{\mathrm{ref}}, \qquad \hat{x}_v^b = \frac{\tilde{x}_v^b}{\|\tilde{x}_v^b\|}. \tag{12}$$

If the projection degenerates, the code switches to a body y-axis reference. The lateral axis is

$$\hat{y}_v^b = \frac{\hat{z}_v^b \times \hat{x}_v^b}{\|\hat{z}_v^b \times \hat{x}_v^b\|}, \qquad \hat{x}_v^b \leftarrow \frac{\hat{y}_v^b \times \hat{z}_v^b}{\|\hat{y}_v^b \times \hat{z}_v^b\|}. \tag{13}$$

The tilt-only mount matrix uses these axes as body-frame columns:

$$C_{bv}^{\mathrm{tilt}} = \begin{bmatrix} \hat{x}_v^b & \hat{y}_v^b & \hat{z}_v^b \end{bmatrix}. \tag{14}$$

A supplied yaw seed is then applied as a vehicle-frame yaw delta:

$$C_{bv,0} = C_{bv}^{\mathrm{tilt}} C_z(\psi_{\mathrm{seed}} - \psi_{\mathrm{tilt}}). \tag{15}$$

After initialization the covariance is reset to tight roll/pitch uncertainty and wide yaw uncertainty.

5 Prediction

Align prediction does not mechanize attitude with gyro integration. The nominal mount is constant:

$$q_{bv,k+1}^- = q_{bv,k}^+. \tag{16}$$

Only the diagonal covariance receives random-walk process noise:

$$P_{ii,k+1}^- = P_{ii,k}^+ + \sigma_{q,i}^2 \Delta t, \qquad i \in \{\phi, \theta, \psi\}. \tag{17}$$

6 Shared Observation Linearization

For a window mean gyroscope ω_b and a body-frame acceleration-like vector a_b, the code predicts vehicle-frame channels as

$$h(q_{bv}) = \begin{bmatrix} \omega_v \\ a_v \end{bmatrix} = \begin{bmatrix} C_{vb}\omega_b \\ C_{vb}a_b \end{bmatrix}, \qquad C_{vb} = C_{bv}^T. \tag{18}$$

With the implemented left small-angle perturbation of q_{bv}, the generated Jacobian rows used by the Kalman updates are

$$H_\omega = C_{vb}\,[\omega_b]_\times\,, \qquad H_a = C_{vb}\,[a_b]_\times\,. \tag{19}$$

Masked rows zero selected columns when an observation should not update yaw or when yaw receives an extra scalar scale.

Scalar and two-row updates use standard linear Kalman equations:

$$r = z - h(\hat{x}), \tag{20}$$

$$S = HPH^T + R, \tag{21}$$

$$K = PH^TS^{-1}, \tag{22}$$

$$\delta x = Kr, \tag{23}$$

$$P^+ \leftarrow (I - KH)P^-. \tag{24}$$

After each update, the small angle is injected into q_{bv} and P is symmetrized. The implementation uses the simplified covariance update above, not the Joseph form.

7 Stationary Gravity Observation

Stationary windows are detected by low gyro norm, accelerometer norm near standard gravity, and near-zero GNSS speed. During such windows, the low-pass gravity vector is updated:

$$g_{b,\mathrm{lp}}^+ = (1-\alpha)g_{b,\mathrm{lp}}^- + \alpha\bar{f}_b. \tag{25}$$

The gravity observation rotates this vector into the vehicle frame:

$$g_v = C_{vb}g_{b,\mathrm{lp}}. \tag{26}$$

For a level stationary vehicle, the constrained channels are

$$g_{v,y} \approx 0, \qquad g_{v,z} \approx -\|g_{b,\mathrm{lp}}\|. \tag{27}$$

The current implementation applies the y and z rows with a roll/pitch mask, so yaw is not directly updated by stationary gravity.

8 GNSS-Derived Horizontal Heading Observation

For consecutive GNSS velocities $v_{n,k-1}$ and $v_{n,k}$,

$$a_n = \frac{v_{n,k} - v_{n,k-1}}{\Delta t}. \tag{28}$$

Using the midpoint horizontal velocity direction

$$\hat{t} = \frac{[v_N, v_E]^T}{\sqrt{v_N^2 + v_E^2}}, \qquad \hat{\ell} = \begin{bmatrix} -\hat{t}_E & \hat{t}_N \end{bmatrix}^T, \tag{29}$$

the GNSS longitudinal and lateral accelerations are

$$a_{\text{long}} = \hat{t}^T a_{n,NE}, \qquad a_{\text{lat}} = \hat{\ell}^T a_{n,NE}. \tag{30}$$

The body accelerometer vector is first stripped of its gravity-axis component. Using the current mount, the body gravity axis is

$$\hat{g}_b = -C_{bv} e_z, \tag{31}$$

and the horizontal body acceleration is

$$a_{b,h} = \bar{f}_b - \hat{g}_b \hat{g}_b^T \bar{f}_b. \tag{32}$$

It is then expressed in the vehicle frame:

$$a_{v,h} = C_{vb} a_{b,h}. \tag{33}$$

When speed and acceleration gates pass, align compares the two horizontal directions by

$$e_\psi = \operatorname{atan2}\left(a_{v,h,x} a_{\text{lat}} - a_{v,h,y} a_{\text{long}}, a_{v,h,x} a_{\text{long}} + a_{v,h,y} a_{\text{lat}}\right). \tag{34}$$

The residual is treated as a vehicle-frame yaw observation:

$$\delta\psi_v = -K_\psi e_\psi, \qquad K_\psi = \frac{P_{\psi\psi}}{P_{\psi\psi} + R_\psi}. \tag{35}$$

The yaw correction is right-injected as $q_{bv}^+ = q_{bv}^- \otimes q_z(\delta\psi_v)$. Roll/yaw and pitch/yaw cross-covariances are zeroed after this scalar yaw update.

9 Turn Consistency and Planar-Gyro Observation

Turn windows require minimum speed, course rate, and lateral acceleration. Before turn-heading observations receive nominal trust, a ring buffer checks

$$a_{\text{lat}} \approx v\,\dot{\chi} \tag{36}$$

for sign and magnitude consistency over several retained windows.

When the turn gyro path is enabled, the body gyro mean is rotated into vehicle coordinates:

$$\omega_v = C_{vb} \bar{\omega}_b. \tag{37}$$

The planar-motion constraint is

$$\omega_{v,x} \approx 0, \qquad \omega_{v,y} \approx 0. \tag{38}$$

By default this update affects roll and pitch. It may update yaw only if the configured turn-gyro yaw scale is positive and turn-heading consistency has passed.

10 Readiness and Known Limitations

The align filter reports coarse alignment ready after yaw has been observed and the one-sigma roll, pitch, and yaw covariances are all below the configured ready gates. Important limitations of the current model are:

- Gravity observes roll and pitch only during stationary windows.
- Horizontal acceleration is projected into a heading/yaw residual; it does not directly solve a complete vector mount least-squares problem.
- The mount process model is a random walk with no vehicle dynamics.
- The yaw update is a scalar vehicle-frame correction, while other mount updates use left-injected body-frame small angles.